

Multiple Choice Questions

Q1. **Answer:** B

Options: A) To store the final output of the entire sequence.; B) To act as a memory that carries information from previous time steps; C) To calculate the loss function for the current word; D) To normalize the input embeddings

Note: Correct option: B - The hidden state acts as a memory that carries information from previous time steps; it is updated at each step and passed forward.

Q2. **Answer:** A

Options: A) To ensure the output values sum to 1, representing a probability distribution; B) To prevent the gradients from exploding; C) To increase the speed of the training process; D) To allow the model to handle variable-length sequences

Note: Correct option: A - Softmax in the final layer converts raw scores (logits) into a valid probability distribution over the vocabulary (values sum to 1).

Q3. **Answer:** B

Options: A) They cannot use embeddings; B) They assume fixed input size and struggle with temporal order; C) They cannot be trained with gradient descent; D) They do not use activation functions

Note: Correct option: B - Feed-forward networks assume a fixed input size and do not naturally model sequential or temporal order; RNNs address this.

Q4. **Answer:** B

Options: A) To update weights based on the loss; B) To calculate the predicted output and compute the loss; C) To unroll the network into a sequence; D) To calculate the gradient for each parameter

Note: Correct option: B - The forward pass computes the predicted output from inputs and current weights, then the loss is computed; weight updates happen in the backward pass.

Q5. **Answer:** B

Options: A) They are randomly re-initialized at every step; B) They are reused across timestamps; C) They increase in value as the sequence progresses; D) Only the output weight V is reused

Note: Correct option: B - In an RNN, the same weight matrices U , V , and W are shared (reused) across all time steps; this is parameter sharing.

Q6. **Answer:** B

Options: A) A deeper network with more representational power; B) A single linear model; C) A recurrent neural network; D) A decision tree

Note: Correct option: B - Composing linear transformations gives another linear transformation; a stack of linear layers is equivalent to a single linear model.

Q7. **Answer:** B

Options: A) The next word given the predicted history; B) The next word given the ground-truth history; C) The current word given itself; D) All words simultaneously

Note: Correct option: B - Under teacher forcing, the model is fed the ground-truth previous token(s) as input when predicting the next word, rather than its own predictions.

Q8. **Answer:** C

Options: A) Only the input at time t ; B) Only the hidden state at time $t - 1$; C) Both the input at time t and the hidden state at time $t - 1$; D) All future inputs

Note: Correct option: C - The RNN recurrence is $h_t = f(x_t, h_{t-1})$; the hidden state at t depends on both the current input and the previous hidden state.

Q9. **Answer:** B

Options: A) $(n_0 \times n_1)$; B) $(n_1 \times n_0)$; C) $(n_0 \times 1)$; D) $(1 \times n_1)$

Note: Correct option: B - For $h = Wx$ with x of shape $(n_0 \times 1)$ and h of shape $(n_1 \times 1)$, matrix multiplication requires W to have shape $(n_1 \times n_0)$.

Q10. **Answer:** B

Options: A) To compute predictions during inference; B) To calculate gradients of the loss with respect to parameters; C) To initialize weights randomly; D) To normalize input data

Note: Correct option: B - Backpropagation computes $\partial L / \partial w$ for each parameter so that gradient-based optimization can update the weights to minimize the loss.

End of Answer Key